# Virtual University of Pakistan

# Bt301

# Introduction to biotechnology

## Full book MCQs

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# BT301 MCQ full book

1. Making of yogurt from milk is the example of biotechnology
    a. **Traditional**
    b. Modern
2. Enzyme Celluloses use for the cleaning of
    a. Protein particle
    **b. Grass marks**
    c. Lipids
3. HUMIRA is an antibody
    a. Diclonal
    **b. Monoclonal**
    c. Triclonal
4. Craig venter present which cells by beating govt concision on HGP
    a. Stem cells
    **b. Ultimate reboot cells**
    c. Adult stem cells
5. Risen bread by using yeast is a process
    **a. Aerobic respiration**
    b. Anaerobic respiration
6. 4000 BC what is use for cheese flavor
    a. Yeast
    **b. Mold**
7. Risen of bread by yeast this process is called
    a. Extraction
    **b. Fermentation**
    c. Cleavage
8. 400 BC food is stored by using
    a. Acids
    b. Bases
    **c. Salt**
9. Leeuwenhoek make how many lenses and microscopes respectively
    a. 100,330
    **b. 200,500**
    c. 200,400
    d. 500,200
10. Enzyme role in the reaction
    a. Higher the activation energy
    **b. Lower the activation energy**
    c. No effect on activation energy
11. DNA present in cell which cell organelle

a. Nucleus
b. Mitochondria
c. Chloroplast
**d. All**

12. MC1R gene controls
    a. U shaped tongue
    b. Curly hairs
    **c. Freckles on face**
    d. Eye color
13. IPO is abbreviation of ?
    **a. INITIAL PUBLIC OFFERING**
    b. INTERNAL PUBLIC OFFERING
    c. INITIAL PUBLIC ORGANIZATION
14. Arthropoitem first drug for?
    a. CANCER
    **b. Anemia**
    c. Arthritics
15. Nerves are covered by sheath called?
    a. Lipids
    **b. Myelin**
    c. protein
16. Restriction enzymes are brought by
    a. Cohen
    **b. Boyer**
17. In which disease which enzyme glucocerebrocidase is deficient?
    **a. Gaucher**
    b. Diabetes
    c. Anemia
18. How many Deoxy-ribonucleotides make the structural units of DNA.
    a. 3
    **b. 4**
    c. 5
19. nucleotides of DNA are joined together through
    a. Peptide linkage
    b. Ester linkage
    **c. Phosphor-diester linkage**
20. DNA base composition change with
    a. Change in environment
    b. Change in food
    **c. Have no effect**
    d. The growing age
21. In DNA, the number of adenosine residues is equal to the number of
    a. Cytosine
    **b. Thymine**

c. Guanine

22. The DNA helix have primary periodicities
    a. 2.4 angstrom
    **b. 3.4 angstrom**
    c. 4.4 angstrom
23. Within two strands of DNA phosphodiester bond run in
    **a. Opposite direction**
    b. Same direction
24. Each ribonucleotide is composed of three components:
    a. A ribose sugar
    b. A Nitrogenous Base
    c. A Phosphoric acid
    **d. All of above**
25. The process of forming mRNA on a DNA template is known as
    a. Translation
    b. Replication
    **c. Transcription**
26. The length of mRNA molecules is variable and it depends on
    a. Length of DNA
    **b. Length of gene**
    c. Length of chromosome
27. tRNA linked with amino acid by
    a. Ionic bond
    **b. Covalent bond**
28. in ribosomes rRNA contain weight about
    a. 50%
    **b. 60%**
    c. 70%
29. result of breaks in helix are
    a. Budges
    b. Internal loops
    **c. Both**
30. Number of stranded amino acids
    a. 19
    **b. 20**
    c. 21
31. Smallest amino acids is
    a. Alanine
    b. Serine
    **c. Glycine**
32. Amino acids are attach with each other by a covalent bond called
    a. Ester bond
    b. Hydrogen bond
    **c. Peptide bond**

33. With the formation of peptide bond between 2 amino acids release a molecule of
    a. Carbon dioxide
    **b. Water**
    c. Hydrogen
34. In primary structure of amino acids all elements are in co-planer but in
    a. Cis configuration
    **b. Trans configuration**
35. The helical twist of the α-helix found in all proteins is
    a. Left handed
    **b. Right handed**
    c. Both
36. The range of $\emptyset, \varphi$ values have a wide range But some values are prohibited because of
    a. Hydrogen bonding
    b. Helical structure
    **c. Steric hindrance**
37. How many amino acid residues in a single turn of helix
    a. 3.4
    b. 2.6
    **c. 3.6**
    d. 3.7
38. adjacent polypeptide chains in a sheet can be
    a. Parallel
    b. Anti-parallel
    **c. Both**
39. A multimer with just a few subunits is called as
    **a. Oligomer**
    b. Protomer
40. An icosahedron is a regular----cornered polyhedron having ----- triangular faces.
    a. 8, 12
    **b. 12, 20**
    c. 20, 12
    d. 12, 8
41. Genetic material in viruses can be in form of
    a. DNA
    b. RNA
    **c. DNA, RNA both**
42. E. coli's chromosome consists of approx. nucleotide pairs.
    a. 5 million
    b. 8 million
    c. 6 million
    **d. 4 million**
43. In E-coli each super coiled loop have average length of
    **a. 10-20 kbp**
    b. 1-2 kbp

c. 7-9 kbp

44. The chromosomes of eukaryotic cells are -----than those of prokaryotes
    a. Larger and less complex
    **b. Larger and more complex**
    c. Small and less complex
    d. Small and more complex
45. DNA with bound histones in the eukaryotes is called as
    a. DNA strand
    b. Chromosome
    **c. Chromatin**
46. Each nucleosome contain -----bp of DNA and --- histones
    a. 100, 8
    **b. 200, 9**
    c. 200, 8
    d. 100, 9
47. Highly condensed chromatin is known as
    **a. Heterochromatin**
    b. Euchromatin
    c. Both
    d. None
48. the study of the entirety of an organism's genes
    a. Genetics
    **b. Genomics**
    c. None
49. Study of proteome is called
    a. Metabolomics
    **b. Proteomics**
    c. Genomics
    d. None
50. By gene sequencing we can understand mechanism of which disease
    a. Infectious diseases
    b. Pathogenic bacteria
    c. Viruses
    **d. All of above**
51. Restriction enzymes are used for
    a. Insertion of vector
    b. Breakdown of proteins
    **c. Digestion of DNA**
    d. None
52. Which one is example of genomic library
    a. AACs
    b. TACs
    **c. BACs**
    d. None

53. YACs is abbreviation of
    a. Yeast artificial colonies
    **b. Yeast artificial chromosomes**
    c. Yeast acidified colonies
    d. None
54. Genomic researches use for
    a. Criminal justices
    b. Civil justices
    c. Establish paternity
    **d. All**
55. Humans have genome size of ------Mb
    a. 2,500
    b. 2,700
    **c. 3,000**
    d. 3,500
56. Viral genome is
    a. Circular
    b. Linear
    **c. Both**
    d. None
57. Lambda contain how many bp in its genome
    a. 47,502 bp
    **b. 48,502 bp**
    c. 48,500 bp
    d. 47,500 bp
58. Influenza virus have which type of nucleic acid and number of genes
    a. ssDNA ,8
    b. dsDNA ,12
    **c. ssRNA ,12**
    d. dsRNA ,8
59. Small organisms carry -----coding density.
    a. Low
    **b. High**
    c. None
60. Over 95% of mitochondrial proteins are encoded in the
    a. Mitochondria
    b. Chloroplast
    c. Endoplasmic reticulum
    **d. Nuclear genome**
61. How many rRNAs present in case of Human Mitochondrial Genome
    a. 12
    b. 37
    **c. 2**
    d. 15

62. Chloroplast genome contain which type of inheritance pattern
    a. Mendelian
    **b. Non- Mendelian**
    c. Both
    d. None
63. Length of Minisatellites
    a. 1-3
    **b. 1-5**
    c. 1-7
    d. 1-13

64. multicellular eukaryotes have  up to genes
    a. 30,000
    **b. 40,000**
    c. 50,000
    d. 60,000
65. Number of genes is correlated to genome size
    a. True
    **b. False**
66. Vertebrate genomes can produce more than one polypeptide per gene because of
    **a. Alternative splicing of RNA**
    b. mRNA
    c. Cellular metabolites
    **d.** Alternative splicing of DNA
67. Introns are
    a. Coding part of DNA
    **b. Non-coding part of DNA**
    c. Both
    d. None
68. Percentage of exons in the human genome
    a. 1.3%
    b. 1.4%
    c. **1.5%**
    d. 1.7%
69. Similarity of man- yeast genome is
    a. 21%
    **b. 23%**
    c. 25%
    d. 27%
70. Distantly related species help us understand events
    **a. Ancient evolutionary**
    b. Recent
    c. Both
71. Bacteria, Archaea, diverged from each other almost
    a. 1 billion years ago

**b. 2 billion years ago**
c. 4 billion years ago
d. 6 billion years ago

72. the FOXP2 gene product turns on genes involved in
    a. Baldness
    **b. Vocalization**
    c. Tumors
    d. All of above
73. prokaryotes genome highly consist of
    **a. Gene**
    b. Transportable elements
    c. Repeats
    d. None
74. which of the following is start codon
    a. GTA
    **b. ATG**
    c. AAG
    d. TTG
75. Upstream region with binding site contain TATA box
    **a. True**
    b. False
76. Polycistronic mRNA encodes for proteins
    a. Few
    **b. Several**
    c. One
    d. Tow
77. Splicing is Removal of introns from the---- molecule
    a. tRNA
    b. rRNA
    **c. mRNA**
    d. All of above
78. Most of prokaryotic genes are without introns and in the are
    a. Monocistronic
    b. Dicistronic
    c. Tricistronic
    **d. Polycistronic**
79. Following are not Nonfunctional copies of genes except
    **a. Pseudo gene**
    b. Duplicated gene
    c. Tandemly repeated gene
    d. Sequencing gene
80. Ethidium bromide is a
    **a.** DNA precipitator
    **b. Loading dye**

c. Digester of DNA
d. None of them

81. Following are example of Non LTR retrotranposons except
    a. LINE
    b. SINE
    **c. ORF**
    d. None
82. Transposon is cut off by
    **a. Transposase**
    b. Proteases
    c. Lipases
83. Retrotransposons, which move by means of an
    a. DNA intermediate
    **b. RNA intermediate**
    c. Both
84. Transposons content in the iris genome is
    a. ~75%
    b. ~85%
    c. ~89%
    **d. ~98%**
85. How many types of transposons in prokaryotes
    a. One
    **b. Two**
    c. Three
86. IS10 is found in R plasmids
    **a. True**
    b. False
87. LTRs stand for
    a. Leading direct Terminal Repeats
    b. Long dominant Terminal Repeats
    **c. Long direct Terminal Repeats**
    d. Long direct Timid Repeats
88. ORF2 encodes an RNA binding protein
    a. True
    **b. False**
89. Which of the following is suitable replicon vehicle
    a. Plasmids
    b. Bacteriophage
    **c. Both**
90. During DNA extraction -----is use for precipitation
    a. Phenol
    **b. Isopropanol**
    c. Isopropyl
    d. EDTA
91. --------is a fragment of DNA or RNA which is used to detect the sequences of target DNA

   a. **Probe**
   b. Primer
   c. None

1. A technique using X-ray film to visualize fragments of molecules that have been radioactively labeled
   a. Blotting
   **b. Autoradiography**
   c. Gel electrophorese
   d. Chromatography
2. Autoradiography is use to analyze the
   a. Length of DNA
   b. Number of DNA
   **c. Both**
   d. None of them
3. A Southern blot is a ----method
   a. Kitchen
   **b. Laboratory**
   c. Both
   d. None
4. After hybridization reaction, membrane is washed and hybridization are detected by
   a. Blotting
   **b. Autoradiography**
   c. Gel electrophorese
   d. Chromatography
5. A northern blot is a method used to detect -----molecules among its mixture
   a. Specific DNA
   **b. Specific RNA**
   c. Specific amino acids
   d. Both
6. In western blotting antibodies are used to detect
   a. DNA fragment
   b. RNA fragment
   **c. Specific antigens**
   d. None
7. To enable the cells to take up circular vector DNA they have to be made
   a. Polar
   b. Linear
   **c. Competent**
   d. Both a and c
8. In electroporation is applied to cells in order to increase permeability of the cell membrane to take DNA
   a. Magnetic field
   **b. Electric field**
   c. Both

9. E. coli is electroporation technique use for introducing
    a. Whole DNA
    b. Fragment of RNA
    **c. Cloned genes**
    d. None of above
10. Animal cells, protoplasts of yeast and plant are susceptible to transformation by
    a. E.coli
    **b. Liposomes**
    c. Bacillus subtilis
11. Amplification of genes done by
    a. Gel electrophoresis
    **b. PCR**
    c. Blotting
12. the phenomenon of restriction and modification elucidated in
    **a. 1960s**
    b. 1970s
    c. 1980s
    d. 1990s
13. Restriction system allow bacteria to monitor the ---- of incoming DNA
    a. Binding
    **b. Origin**
    c. Recombining site
    d. None
14. For phage ʎ restriction/ modification is studded on which of the following strain
    a. E. coli C
    b. E. coli K
    c. E. coli D
    **d. E. coli C and E. coli K**
15. Methylation of certain bases is the process happened in
    a. Restriction
    **b. Modification**
    c. Recombination
    d. All
16. How many type of R-M system are known
    a. 1
    b. 2
    c. 3
    **d. 4**
17. In type 1 The active enzyme consists of --restriction subunit,-- modification subunit and --recognition subunit
    a. 2, 3, 2
    b. 1, 2, 3,
    **c. 2, 2, 1**
    d. 3, 2, 1

18. Most of the useful R-M system is
    a. Type I
    **b. Type II**
    c. Type III
19. Type ---enzymes recognize a defined sequence and cut within it
    a. Type I
    **b. Type II**
    c. Type III
20. Restriction occur distant from the recognition site this statement is true for
    a. Type I
    b. Type II
    c. Type III
    **d. Type IIs**
21. R. HindIII in this name R indicates
    a. Modification-methylase
    b. Specie name
    **c. Endonuclease**
    d. None
22. AAGCTT is recognition site for
    a. HaeIII
    **b. HindIII**
    c. HindII
    d. HamHI
23. The number and size of fragments done by restriction enzyme depend on
    a. Origin site of DNA
    b. Recognition site of DNA
    **c. Target site of DNA**
    d. Binding site of DNA
24. Six base recognition site occurs every
    a. $4^4$ bp
    **b. $4^6$ bp**
    c. $4^8$ bp
25. ------ the enzyme that degrade DNA
    **a. Nucleases**
    b. DNA polymerase
    c. Reverse transcriptases
    d. DNA ligases
26. An enzyme that creates a phosphodiester bond
    a. Nucleases
    b. DNA polymerase
    c. Reverse transcriptases
    **d. DNA ligases**
27. Mainly----- methods are used for joining DNA in vitro
    a. 1

b. 2
**c. 3**
d. 4

28. ------can repair the nicks produced after the association of cohesive ends of DNA strands
    a. Nucleases
    b. DNA polymerase
    c. Reverse transcriptases
    **d. DNA ligases**
29. T4 DNA ligase has been used to joint ----DNA molecules
    a. Cohesive ends
    **b. Blunt-ended**
    c. Single stranded nicks
30. The main difference between linker and adaptor is that former having
    a. Cohesive ends
    **b. Blunt-ended**
    c. Single stranded nicks
31. Eukaryotic mRNA can be cloned in vector after converting it to
    a. DNA
    b. tRNA
    **c. cDNA**
32. Naturally occurring bacterial plasmids range in size from
    a. 500 to 40,000 bp
    **b. 5,000 to 400,000 bp**
    c. 4,000 to 500,000 bp
    d. 3,000 to 400,000 bp
33. Open circle or OC DNA have
    a. Both strand intact
    **b. One strand intact**
    c. Completely closed
34. -----may be a mutagen, a carcinogen, or a teratogen
    a. Phenol
    b. Isopropanol
    **c. Ethidium bromide**
    d. Oxygen
35. Which plasmid carry set of tra gene
    **a. Conjugated**
    b. Non- conjugated
    c. Both
36. Host range of plasmid is determined by
    a. Binding site
    **b. Ori region**
    c. Tra gene
    d. None
37. Naturally occurring plasmids are stably maintained because they contain

a. Ori region
b. Tra gene
**c. Par region**
d. None

38. --------of plasmids may arise by deletions or rearrangements of DNA
    a. Incompatibility
    **b. Structural instability**
    c. segregative stability
    d. Partitioning
39. Single sites for number of restriction endonucleases is property of --- cloning vehicle
    a. Naturally occurring
    **b. Ideal**
    c. Every
    d. Synthetic
40. Plasmids which were not constructed in vitro for the sole purpose of cloning are called plasmid
    a. **Natural**
    **b.** Ideal
    c. Every
    d. Synthetic
41. pSC101 Contain
    a. Immunity to colicin
    b. Ampicillin resistance
    c. **Tetracycline resistance**
    d. Colicin E1 production

# For final term

For final term

- pSC101 contain ------resistance
  Ampicillin
  **Tetracycline**
  None
- pBR322 is an example of -----constructed cloning vehicle
  In vivo
  **In vitro**
  Artificially
- pBR322 contains the resistance gene against
  Ampicillin
  Tetracycline
  **Both**
  None

- in pBR322 replicon element is
  pBR11
  **pMB1**
  RSf2124
  PstI
- pBR322 have -----target sites for different restriction enzymes
  30
  20
  **40**
  50
- pBR322 have high copy number and -----weight
  High
  **Low**
- pBR325 encodes ----resistance in addition to those resistance have psc101 plasmid.
  Ampicillin
  **Chloramphenicol**
  Tetracycline
- pAT153 another derivative of
  pBR325
  **pBR322**
  psc101
- plasmids which loss during cell division is called
  High copy number plasmid
  Low copy number plasmid
  **Runaway plasmid**
  Lambda phage vector
- Wild type is not suitable as a cloning vector because it has ----restriction enzymes sites
  Very low number of
  **Too many**
  A few
- Charon4A is a replacement vector is this statement
  **True**
  False
- Charon 60 is a replacement vector is this statement
  True
  **False**
- Charon 60 has ----- restriction site
  **1**
  2
  3
  Many
- Charon 4A and 16 are both derivatives of ------
  Bacteriophage
  **Lambda**

Ecoli

- Charon 60 can be cloned
  2 kbp
  3kbp
  4kbp
  **5kbp**
- lacZ gene encode for enzyme
  PAL
  4CA
  **βGal**
  PPR
- first step in the cloning is
  Cutting vector
  **Isolation of DNA**
  Insertion of vector
  Infection
- in case phage-λ Packaging yields about ----- of vector DNA
  $10^4$ plaques/ µg
  $10^5$ plaques/ µg
  **$10^6$ plaques/ µg**
- At each cos site nicks are introduce ----- far apart on opposite strand of DNA
  10bp
  13bp
  **12bp**
  16bp
- Cos sits are
  Cloning site
  Cooperative site
  **Cohesive site**
  Capsulation site
- Gene D is include in the --- process
  **Capsulation**
  Ligation
  Termination
  Packaging
- M13 is a filamentous bacteriophage with-----
  dsDNA
  **ssDNA**
  dsRNA
  ssRNA
- Which of the following vector replicate without killing its host
  Phage-λ
  **M13**
  Mp18

Cosmid

- M13 can clone DNA of variable length about
  3kpb
  4kbp
  **5kbp**
  6kbp
- In x gal containing medium blue colonies represent the cloned DNA colonies, statement is
  True
  **False**
- Cosmid can clone around ------length of DNA
  **45kbp**
  37kbp
  53kbp
  200kbp
- Modified cosmid vector produced by ----- in 1981
  Watson and crick
  Ish-Horowicz and Burke
  Edward Southern
  **Bates and Swift**
- A modified procedure for cosmid is derived by
  Watson and crick
  **Ish-Horowicz and Burke**
  Edward Southern
  Bates and Swift
- Cosmid c2XB is derived by
  Watson and crick
  Ish-Horowicz and Burke
  Edward Southern
  **Bates and Swift**
- -----vectors are combination of plasmid and ʎ phage sequences
  Lambda
  **Phasmid**
  Cosmid
- BACs can clone ----------length of DNA fragment
  5kbp
  150-200kbp
  **100-300 kbp**
  60kbp
- YACs vectors replicate in ------like normal chromosome
  Bacteria
  **Yeast**
  Fungi
  Mammalian cell
- YACs can clone ----------length of DNA fragment

5kbp
**150-200kbp**
100-300 kbp
60kbp

- Vectors that can replicate and are stably maintained in two or more unrelated host organisms are called
  YACs
  **Shuttle vectors**
  BACs
  Cosmid vector
- Expression vector are used to control
  Infection
  **Expression**
  Translation
  Insertion
- In Ecoli p1 phage can clone -----length of DNA
  10-15 kb
  2-25 kb
  35-45 kb
  **70-100 kb**
- The second strategy is to selectively amplify the target sequence directly from the source DNA by using
  Different vectors
  **PCR**
  Gel electrophoresis
- ---- represent the genomic makeup of an organism
  cDNA library
  RNA sequence
  **Genomic library**
- Genome size of human is
  **$2.8x10^6$**
  $3x10^6$
  $1.25x10^5$
- Drosophila melanogaster have size of DNA $1.8x10^5$ this statement is
  **True**
  False
- Average fragment size can be controlled by----------
  Restriction endonucleases
  **Mechanical shearing**
  Random cutting
- High molecular weight genomic DNA is digested with-------
  BamH1
  ***Sau*3AI**
  Sal1

PmB1

- For construction of genomic libraries ----vectors are required
  **High capacity cloning**
  Low capacity cloning
  Moderate capacity cloning
- ------is prepared by reverse-transcribing cellular RNA
  DNA
  **cDNA**
  Proteins
  Amino acids
- Which lack the non-coding sequence
  DNA
  rRNA
  **cDNA**
- Introns are rare in bacteria but occur in genes of higher
  **Eukaryotes**
  Prokaryotes
- cDNA library is representative of the ---population from which it was derived
  DNA
  **RNA**
  Proteins
- A serious disadvantage of the hairpin method is that cleavage with S1 nuclease results in the loss of sequences at the
  3' end of the clone
  **5' end of the clone**
- correct orientation of inserted cDNA can be achieved by------
  Hairpin method
  **Self-priming method**
  Sequencing
  Reverse transcriptase method
- -----followed by the PCR (RTPCR) leads to the amplification of RNA sequences in cDNA form
  Genomic library
  **Reverse transcription**
  Splicing
  Digestion of DNA
- Major screening strategies involve Genetic methods
  Sequence-dependent screening
  Screening expression libraries
  **All of above**
- All useful vector molecules carry a selectable ---------or property
  Nutritional marker
  **Genetic marker**
  Drug resistance
- In phage vectors, ------is itself the selected property

$AP^R$
$TC^R$
**Plaque formation**
$CH^R$

- Nucleic acid ----------is the most commonly used method of library screening
  Amplification
  Sequencing
  **Hybridization**
  Screening
- --------- (1975) developed a screening procedure to detect DNA sequences in transformed colonies
  Ish-Horowicz and Burke
  Edward Southern
  Bates and Swift
  **Grunstein and Hogness**
- The results of the hybridization can be monitored by
  PCR
  **Autoradiography**
  Gel documentation
  Centrifuge
- Benton and Davis (1977) devised a method called
  Hybridization
  **Plaque lift**
  Sequencing
  Screening
- In plaque lift techniques there is direct contact between plaque and filter this statement is
  False
  **True**
- A great advantage of hybridization for library screening is that it is extremely
  Important
  **Versatile**
  Difficult
  Unique
- A cloned genomic fragment must be found as a ------point for the walk
  Mid
  End
  **Starting**
  Selectable
- In chromosome walking, human genome starting point may be a ----
  CAPs
  SNPs
  **RFLP**
  AFLP

- One----- of chromosome walking is the requirement that each DNA segment used is not repeated elsewhere in the genome
  Benefit
  **Drawback**
  Characteristics
- In chromosome jumping, the DNA of interest is identified, cut into fragments with restriction enzymes and
  Elongate
  **Circularized**
  Break down
  Straighten
- Cloned DNAs from the closure sites make up a jumping library is this statement
  False
  **True**
- The PCR is widely used to isolate specific DNA sequences from-----genomic DNA or cDNA
  Cloned
  **Uncloned**
  Gel treated
- To isolate specific clone, PCR is carried out with
  Simple primers
  Unusual primers
  Special primers
  **Gene-specific primers**
- A degenerate primer is a mixture of primers, all of similar sequences but with variations at --- positions
  One
  Two
  **One or more**
  Several
- If DNA library is established using expression vectors, each individual clone can be expressed to yield a
  Monopeptide
  **Polypeptide**
  Dipeptide
- Immunochemical screening involves the use of
  **Antibodies**
  Antigen
  Nucleic acid
  RNA
- The molecular target for recognition is generally an
  Antibody
  **Epitope**
  Probe
  Primer

- polyvinyl and that IgG antibodies can be readily labelled with
  **$I^{125}$**
  $I^{108}$
  $I^{127}$
  $I^{129}$
- it is much more convenient to use--- because they have high capacity and efficiency to in vitro packaging
  Bacteriophage simple insertion vector
  Bacteriophage Ecoli 1 insertion vector
  **Bacteriophage-ʎ insertion vector**
  Bacteriophage-ʎ deletion vectors
- Screening is carried out without using an antibody, by incubating the membranes with radiolabelled ----------
  Ss DNA probe
  **ds DNA probe**
  Ss RNA probe
- -----is the process by which a particular DNA sequence compensates for a missing function in a mutant cell and thus restores the wild type phenotype
  **Functional complementation**
  Functional analysis
  Functional genomics
  Non Functional complementation
- Functional complementation is also possible in
  **Transgenic animals and plants**
  Transgenic animals
  Transgenic plants
  Transgenic bacteria
- Functional complementation used for complementation in -----to isolate the Shaker - 2 gene
  Transgenic cat
  **Transgenic mice**
  Transgenic bacteria
- -----depends upon transcription of appropriate gene, efficient translation of mRNA and in many cases, posttranslational processing and compartmentalization of nascent polypeptide
  Synthesis of functional DNA
  Synthesis of functional RNA
  **Synthesis of functional protein**
- Gram-negative bacteria such as E. coli have a
  Complex wall-membrane mitochondria
  Simple wall-membrane
  **Complex wall-membrane**
  Difficult wall-membrane
- The bacterial inner membrane, periplasmic space and outer membrane all contain proteins found in the cytoplasm
  True

**False**

- ABC pathway of protein export and type II secretion pathway are use for
  Chromosome trafficking
  DNA trafficking
  RNA trafficking
  **Protein trafficking**
- In E.coli foreign protein are
  Stable
  **Instable**
  Dissolved
  Submerged
- In the case of somatostatin, degradation was prevented by producing a ------consisting of somatostatin and β-galactosidase
  Simple protein
  **Fused protein**
  Complex protein
  Hybrid DNA
- Expression from a strong promoter can represent ----of cloned gene product of toral cell protein
  30-40%
  10-40%
  **20-40%**
  20-30%
- Complementarity of Shine-Dalgarno (S-D) sequence with 16s rRNA can affect the
  Rate of transcription
  **Rate of translation**
  Rate of enzyme
  Rate of cell cycle
- Composition of triplet immediately preceding the AUG ----also affects the efficiency of translation
  **Start codon**
  Stop codon
  Termination site
- The rate of synthesis of a particular protein will depend on the ---------in the cell
  Steady-state of RNA
  Steady-state of DNA
  **Steady-state of mRNA**
  Mobile-state of mRNA
- ---------usually proceeds by a combination of endonuclease and 3'exonuclease attack
  Synthesis of mRNA
  **Degradation of mRNA**
  Degradation of DNA
  Degradation of protein
- One way of increasing the expression of a cloned gene is to increase the
  Temperature

**Plasmid copy number**
Enzyme
pH

- The loss of plasmids due to defective partitioning is called
  Segregative stability
  **Segregative instability**
  Non-Segregative stability
  Non-Segregative instability
- Structural instability of plasmids may arise by deletions or rearrangements of
  RNA
  mRNA
  **DNA**
  rRNA
- Techniques for DNA sequencing became available in the
  Early 1970s
  **Late 1970s**
  Early 198s
  Late 1980s
- Maxam-Gilbert method for DNA sequencing makes use of chemical reagents to bring about ------ of the DNA
  Base-specific addition
  **Base-specific cleavage**
  Digestion
  Rearrangement
- DNA synthesis is carried out in the presence of the four deoxynucleoside triphospahtes, one or more of which is labelled with ----- and in four separate incubation mixes
  $N^{15}$
  **$P^{32}$**
  $N^{14}$
  $P^{35}$
- The combination of chain terminator and M13 vectors to produce -----is very powerful
  dsDNA
  **ssDNA**
  dsRNA
  ssRNA
- Sequences that were read beyond 4oo bp contained an average of ---error
  2.3%
  **3.2%**
  3.4%
  4.4%
- Sequences that were read less than 4oo bp contained an average of ---error
  2.3%
  **2.8%**
  3.2%

4.4%

- --------is a process to change the genetic information of an organism
  Hybridization
  Autoradiography
  **Mutagenesis**
  Genetic library
- Mutagenesis sources
  Occur naturally
  Exposure to mutagens
  Induced in laboratory
  **All of above**
- ----different methods of site-directed mutagenesis
  2
  **3**
  4
  5
- A synthetic DNA fragment containing the desired mutant sequence is used to replace the corresponding sequence in the wild-type gene
  **Cassette mutagenesis**
  Primer extension
  Procedures based on PCR
- In -----method, the product of the first PCR is used for the second PCR
  Site directed PCR
  **Megaprimer**
  Basic PCR
- The ----of a PCR based mutagenic protocol is that the desired mutation is obtained with 100% efficiency
  **Advantage**
  Disadvantage

  None
- PCR can also be used to generate DNA fragments for gene cloning by using ---with specific restriction sites
  Probes
  **Primers**
  Both
  None
- Nucleotides are building block of
  Proteins
  Lipids
  **DNA**
  All of above
- Taq polymerase is a ----DNA
  **Thermostable**

Thermoinstable
Binding resistant

- ----------is a short strand of oligonucleotide that serves as a starting point for DNA synthesis
  Probes
  **Primers**
  Both
- --------was designed to amplify RNA sequences (mainly mRNA) through synthesis of cDNA by reverse transcriptase
  **RT-PCR**
  q PCR
  Nested PCR
  Inverse PCR
- It is used to measure starting amounts of DNA, cDNA, or RNA
  RT-PCR
  **q PCR**
  Nested PCR
  Inverse PCR
- In --------two sets of primers are used in two successive PCRs
  RT-PCR
  q PCR
  **Nested PCR**
  Inverse PCR
- ----------is used to identify the flanking sequences around genomic inserts
  RT-PCR
  q PCR
  Nested PCR
  **Inverse PCR**
- Multiplex PCR is used to amplify several different DNA sequences simultaneously
  **True**
  False
- ---------is a type of PCR reaction but the segments of DNA that are amplified are random
  AFLP
  **RAPD**
  SNPs
  RFLP
- AFLPs-Amplified fragment length polymorphisms are differences in restriction fragments length caused by
  AFLP
  RAPD
  **SNPs**
  RFLP
- PCR has applications in the fields of life sciences
  Genetic engineering
  Medical

Forensic
**All of above**
Both a b

- PCR has been used in gene cloning and ------of genomic libraries
  Sequencing
  **Screening**
  Construction
  Degradation
- DNA profiling or DNA fingerprinting have --------meaning
  Different
  **Same**
- DNA profiling or DNA fingerprinting is a forensic technique used to identify individuals by characteristics of their..
  Protein
  RNA
  **DNA**
  cDNA Genomic library
- Assigning of a specific gene to particular region of a chromosome and determining the location of and relative distances between genes on the chromosome is called
  Chromosome jumping
  Chromosome walking
  **Chromosome mapping**
- Any identifiable feature of the genome cannot serve as a marker in mapping, but specific
  True
  **False**
- The landmarks on genome map might include
  Short DNA sequences
  Regulatory sites
  Genes themselves
  **All of above**
- Genetic linkage maps illustrate the order of genes or genetic markers on a chromosome and the relative distances between those genes
  DNA finger printing
  Physical mapping
  **Genetic linkage mapping**
- ----------give the DNA base pair distances from one landmark to another
  DNA finger printing
  **Physical mapping**
  Genetic linkage mapping
- ----------exploits variations in homologous DNA sequences
  AFLP
  RAPD
  SNPs
  **RFLP**

- Botstein et al. (1980) were the first to recognize that ----that target RFLPs can be used for mapping
  **DNA probes**
  DNA primers
  RNA primers
- Sequence-tagged sites (STS) are more convenient markers than RFLPs because they do not use RT-PCR
  **Southern blotting**
  Restriction enzyme
  DNA polymerase
- SNPs are -----base-pair positions in genomic DNA at which different sequence alternatives (alleles) exist in a population
  **Single**
  Double
  Triple
  Complementry
- Amplified fragment length polymorphisms (AFLP) is a diagnostic fingerprinting technique that detects genomic
  Amplified fragment
  **Restriction fragments**
  Complemenrtry fragment
- FISH is a cytogenetic method that used ----probes
  Scanning
  **Fluorescent**
  Radio active
- RH mapping used X-ray breakage of chromosomes to determine the distances between ---as well as their order on the chromosome
  DNA probes
  DNA primers
  **DNA markers**
- Happy mapping is an ------technique
  In vivo
  **In vitro**
  None
- The process of determining the exact order of nucleotides within a DNA or RNA molecule is called
  Screening
  **Sequencing**
  Mapping
- Shotgun sequencing is used for sequencing -------DNA strands
  Short
  **Long**
  Fragmented
  Digested

- --------- a map of each chromosome of the genome is made before the DNA is split up into fragments for sequencing
  Chromosome sequencing
  **In clone-by-clone sequencing**
  Physical mapping
  Genetic linkage mapping
- Orthologs are homologous genes in different organisms that encode proteins with the
  Different functions
  **Same function**
  Non functional
- Genome of bacteria may be of variable in size. For example, it may vary from
  0.39Mb to 8.1 Mb
  **0.49 Mb to 9.1 Mb**
  0.49Mb to 1.9Mb
- Mitochondrial genomes exhibit an amazing
  Genomic diversity
  **Structural diversity**
  Functional diversity
  Phenotypic diversity
- ----are used to study the expression of many genes at once
  Comparative genomics
  **Microarrays**
  Filamentous bodies
- Which is the DNA array used in expression analysis
  Spotted DNA arrays
  Printed oligonucleotide chips
  **Both**
  None
- The predominate application of DNA microarray has been to measure
  **Gene expression levels**
  Gene synthesis level
  Genetic difference
- Microarrays have been widely used as ----------genotyping platforms
  AFLP
  RAPD
  **SNP**
  RFLP
- Phage display is the technology that allows expression of exogenous -----on the surface of phage particles
  Monopeptide
  Dipeptide
  **Polypeptides**
  Nucleotide

- The most common screening method is based on enriching the phage clones with binding affinity for the target by a process called
  Epitope
  **Biopanning**
  Microarrays
- Applications of phage display includes
  Determination of protein to protein interactions,
  To determine enzyme specificity
  To generate target specific antibodies
  Both a b
  **All of above**
- A genetic technique in which one of an organism's gene is made inoperative that is simply called
  Knocked in
  **Knocked out**
  Knocked down
- ---- refers to a gene manipulation method that involves the insertion of a protein coding cDNA sequence at a particular locus in an organism's chromosome
  **Knocked in**
  Knocked out
  Knocked down
- Small interfering RNA (siRNA) is the most commonly used (RNAi) tool for inducing ----------coding genes
  Long-term silencing of protein
  **Short-term silencing of protein**
  Permanent silencing of protein